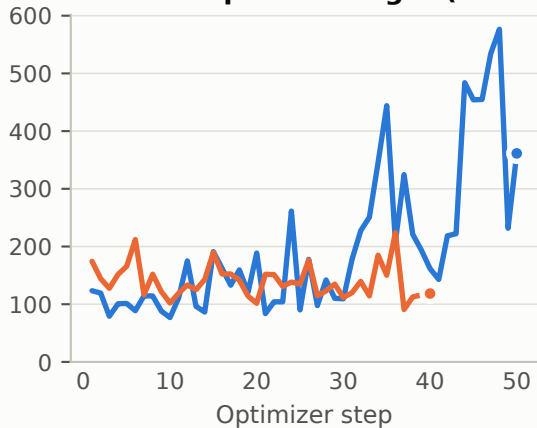
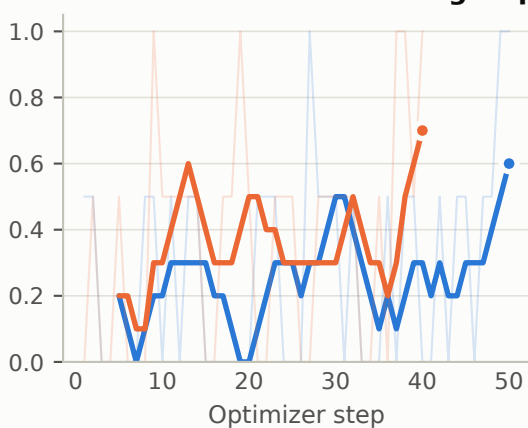


GRPO training dynamics: run 1 drifts to long completions; run 2 stays stable

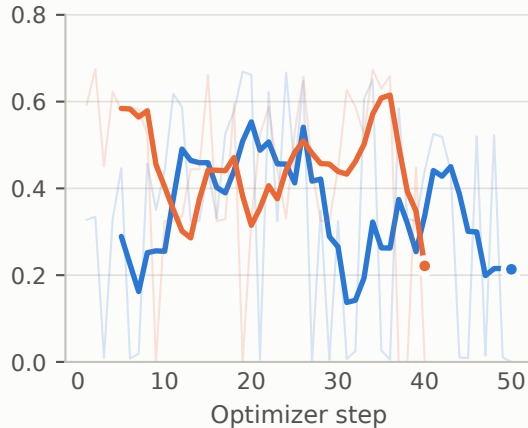
Mean completion length (tokens)



Fraction of zero-variance groups



Reward std within batch



— GRPO run 1 (group-std, reward v1, lr $2e-5$)

— GRPO run 2 (Dr. GRPO, reward v2, KL 0.02, lr $1e-5$)

Right two panels: 5-step moving average (bold) over raw per-step values (faint). Reward scales differ between runs (v1 vs v2) and are not compared directly.